Pediatric analgesia

Earlier preoperative percutaneous intercostal cryoanalgesia improves recovery after pectus excavatum surgery

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Abstract

Background and aims Minimally invasive repair of pectus excavatum is associated with intense postoperative pain. We aimed to evaluate the effectiveness of percutaneous intercostal cryoanalgesia according to the different timing of its preoperative application. Outcome variables included pain, drug consumption, time to mobilization, and hospital length of stay.

Methods Retrospective data analyzed from a single pediatric institution, involving a series of consecutive patients who underwent pectus surgery. Group 1: 18 patients received thoracic epidural analgesia. Group 2: 11 patients underwent cryoanalgesia on the same day of surgery and epidural analgesia as a bridge to cryoanalgesia onset. Group 3: 13 patients underwent earlier preoperative intercostal cryoanalgesia with a median of 6 days (IQR 3–6) before surgery. All groups received rescue systemic analgesia.

Results Pain scores were similar among groups. The incidence of severe pain decreased in Group 3 versus Group 1 (38% vs 78%) ($p<0.05$). Ropivacaine epidural cumulative dose (mg/kg) decreased in Group 2 versus Group 1: 16 (IQR 12–23) versus 27 (IQR 22–35) ($p<0.01$). Cumulative oral morphine equivalent dose (mg/kg) decreased among groups: 12 (IQR 9.3–17), 8 (IQR 4.2–12), and 0.2 (IQR 0.0–0.3) ($p<0.001$) for Groups 1–3, respectively. Length of stay (days) also decreased among Groups 1–3, respectively: 7 (IQR 6–7), 6 (IQR 5–7), and 4 (IQR 4–5) ($p<0.001$).

Conclusions This exploratory observational study suggests that the timing of cryoanalgesia may play a crucial role in its efficacy, with earlier placement being associated with improvements in pain, opioid utilization, and length of stay.

Data availability statement

Data are available upon reasonable request. Data should be available upon reasonable request to the institutional DPO (data protection officer).

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Footnotes

Contributors: SM is the principal investigator; conceived, designed and executed the study; collected and interpreted the data; and drafted the manuscript and the revisions. AB helped conduct the study. IP helped to conceive the study, collect and interpret the data. LC analyzed the data. MCa helped collect the data. MCh helped promote and execute the study. EB is the guarantor of the study and helped conduct the study, having had access to all the work phases.

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